



# Data Osmosis: Rapid Absorption of Dataset Knowledge through GPT-Enhanced Documentation

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## Formation Bio

# Agenda

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- The Solution: GPTs as a Catalyst in Pharma
- Diffusion in Action: Implementing GPT-Enhanced Processes
- Tailoring the Membrane: Custom GPTs in Pharma
- Boosting Osmosis: GPTs for Dataset Integration
- Overcoming Saturation: Advanced GPT Techniques
- Future Reactions: Conclusion and Next Steps
- Q&A



# **The Solution: GPTs as a Catalyst in Pharma**

# Planning an AI Experiment

- Generative Pre-trained Transformer (GPT) models present the opportunity to augment workflows to overcome common challenges and inefficiencies if rolled out within an organization effectively
- Implementing enterprise versions of these tools in pharma requires careful planning, risk assessment, policy development, training, and support

**Get things set-up**

**Drive adoption**

**Review use + value**

# ChatGPT Overview



| Tool                                                      | Context of use                                                                                                                                                           | Choose this tool when...                                                                                                                   | Data retention                                                                                                   | Example(s)                                                                                                                                    |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ChatGPT</b>                                            | Conversation with the internet of information                                                                                                                            | There is no need to upload information (you could ask the general public)                                                                  | N/A                                                                                                              | <ul style="list-style-type: none"><li>→ Questions about FDA guidelines</li><li>→ First draft of R code</li></ul>                              |
| <b>Enterprise Custom GPT</b>                              | A tool trained on tasks and topics by combining specific instructions, knowledge, and access to internal systems                                                         | This is a repeatable task or topic trained on specific information (you could ask the intern)                                              | Uploaded data is transferred to OpenAI and retained and/or used to train the model as defined in the agreement   | <ul style="list-style-type: none"><li>→ Questions about packages</li><li>→ Generate potential data checks or analyses for a dataset</li></ul> |
| <b>Assistants Application Programming Interface (API)</b> | Used by an internal applications to create an assistant that can have access to specific information and instructions and respond to user queries within the application | You need a bot within an application such as Teams trained on specific information (you could ask a previous intern hired as an assistant) | Data is transferred to OpenAI via an API and retained and/or used to train the model as defined in the agreement | <ul style="list-style-type: none"><li>→ Bot that answers questions about company HR policies you can ask a Teams channel</li></ul>            |



# Diffusion in Action: Implementing GPT-Enhanced Processes

# Equipment Set-up

- The most crucial and often time-consuming steps in creating custom GPTs is curating the specific knowledge for the GPT to train on and creating effective prompt instructions

## Knowledge

- Keep the number of files used to 20 or less (system limitation currently 20 files)
- You can merge multiple files as long as size kept to 512MB and use ChatGPT to help merge

**Tip:** If you upload "Knowledge", "Code Interpreter" must be selected for your GPT to use that information

## Instructions

- Provide relevant context and detailed, specific instructions
- Don't use phrases in the negative in your instructions

**Tip:** Saying something like "Please include colorful illustrations in your response" is more straightforward for an AI to interpret and act on than saying "Don't leave out colorful illustrations in your response."

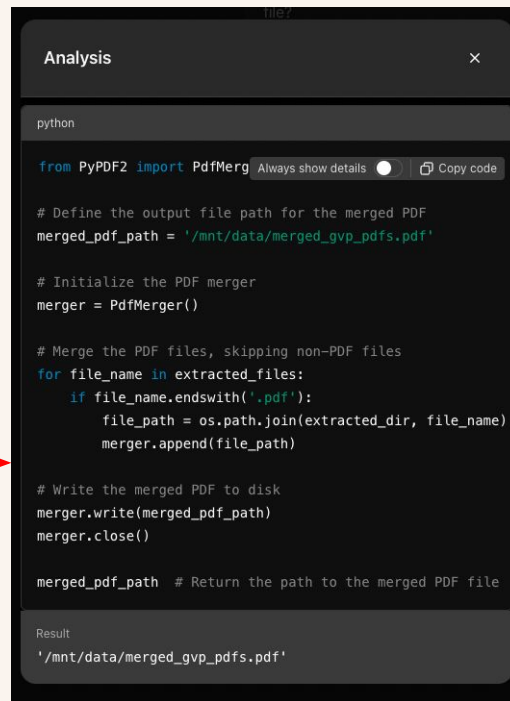
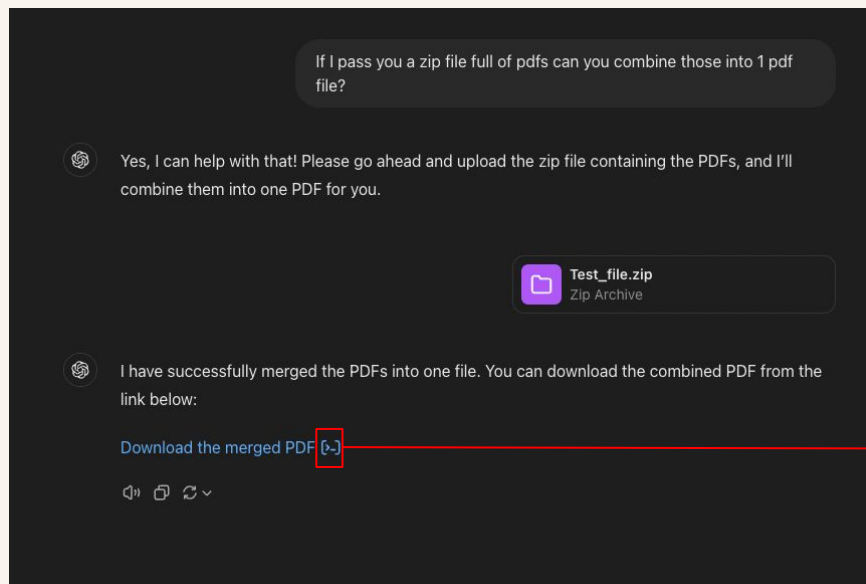
# ChatGPT Data Limits

- **ChatGPT Conversations and Custom GPTs support upload and export of all common file extensions for text files, spreadsheets, presentations, and documents.**
- **How many files can I analyze at once?**
  - Up to 10 files can be uploaded to a given conversation
  - Up to 20 files can be attached to a GPT as Knowledge (ChatGPT can interact with these files if the Code Interpreter capability is enabled at the GPT level)
- **How much data can I analyze?**
  - 512 MB per file. For CSV files or spreadsheets, the file size cannot exceed approximately 50MB, depending on the size of each row.





# Using GPT to augment manual workflows: Combining PDFs



Navigate around document limits by passing GPT a single zip file

Download a merged file in seconds and the python code running in the background



# Tailoring the Membrane: Custom GPTs in Pharma

# Data Governance

If your tools require training on specific information, consider if this information is regulated, private, or public.

## ● Regulated data (PHI, PII, Clinical Trial Data, etc.)

- Must not be used directly in the ChatGPT enterprise interface
- AI tools cannot be used to make GxP decisions without further vetting and validation

## ● Private data (de-identified personal, restricted, and proprietary)

- Company-approved enterprise AI tools
- Consider who has access to this information when sharing Custom GPTs. Default access permissions are creator plus invite-only

● Non-sensitive public data can be used in other approved and publicly available AI tools (e.g. Claude, Gemini, etc.)

# Risk Considerations

## Governance

Discrepancy over the data definition of a KPI (e.g. how is “profit margin” defined?)

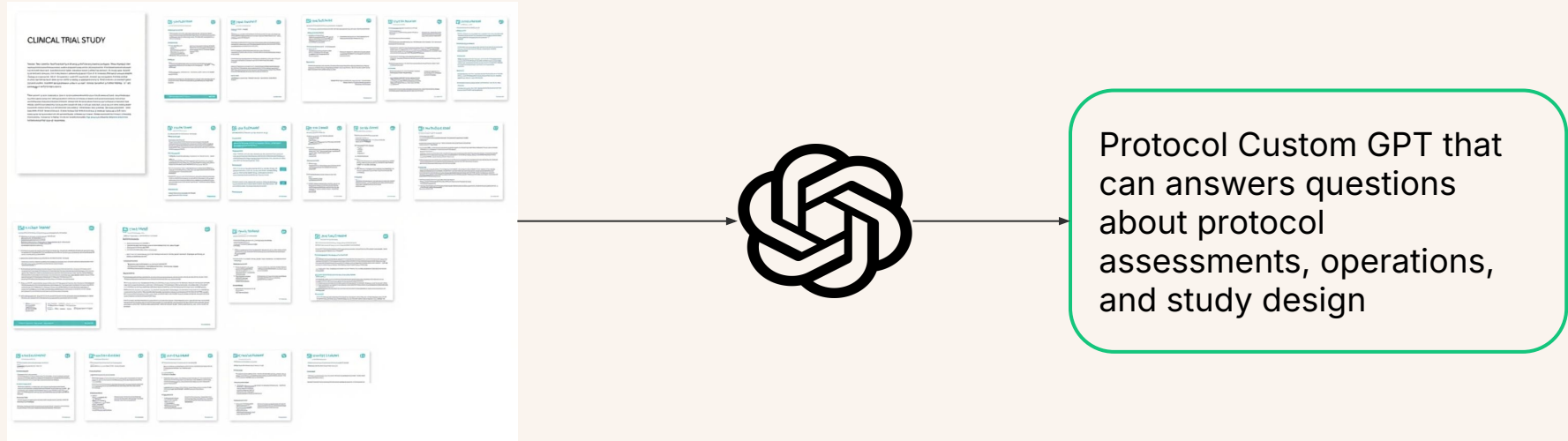
## Hallucinations

ChatGPT may make an error in its code (even if it has the correct data definition)

## Human in Loop

A human must be in the loop to kick off a ChatGPT prompt (not programmatic)

# Case Study: Protocol-Specific Custom GPTs

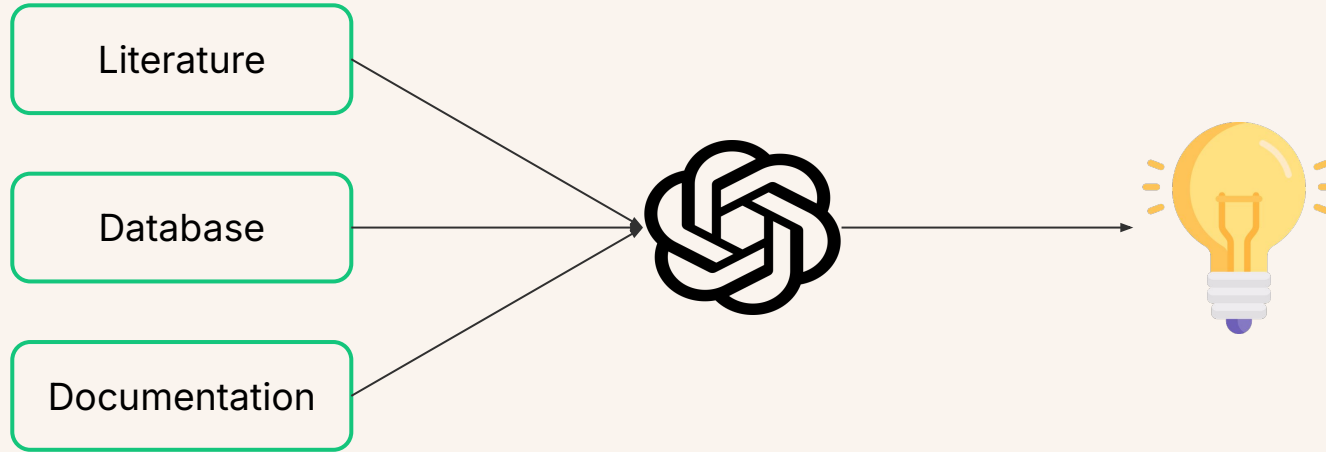




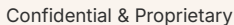
# Boosting Osmosis: GPTs for Dataset Integration

# Using GPTs to reach a data equilibrium

- Publicly available databases present an opportunity to leverage GPTs to collate, understand, and interpret these datasets to speed up analysis



## Formation Bio





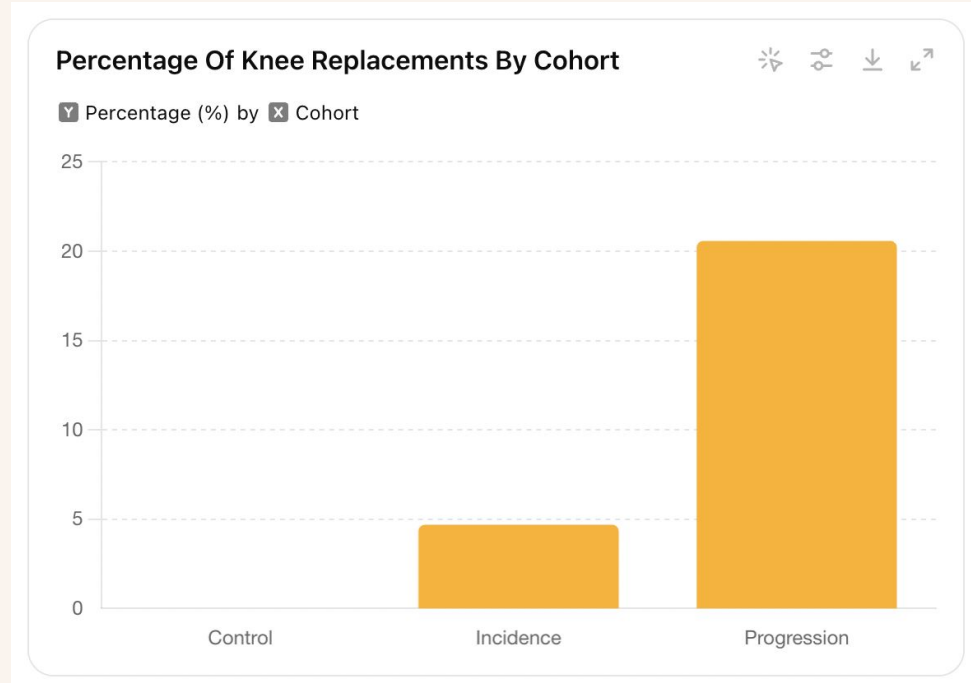


# Overcoming Saturation: Advanced GPT Techniques

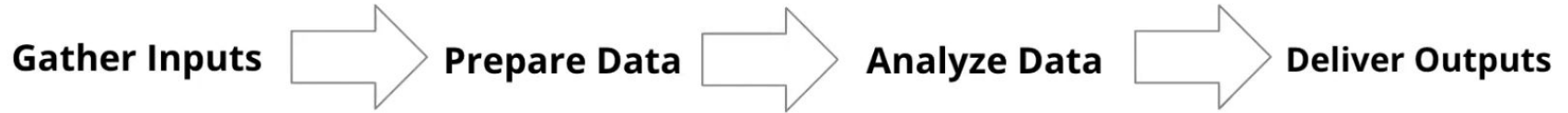
# Enterprise Data Analyst GPT

- Collaborate with your own AI analyst to get from data to insights faster
- Capabilities:
  - Chat with your data
  - Get proactive insights and suggestions
  - Create customizable, presentation-ready charts
- ChatGPT Data Analyst can analyze data uploaded in a variety of file formats, including:
  - Excel (.xls / .xlsx)
  - Comma-separated values (.csv)
  - PDF (.pdf)
  - JSON

**Note:** This GPT has Enterprise-grade data privacy and security so the model is not trained on your data but as always be sure to reference company data governance guidelines before uploading any data.



# GPT Data Analysis Process



Actions

Data  
Analysis

Data  
Analysis

Actions

- Write SQL
- Execute SQL

- Describe data
- Update columns
- Anomaly detection

- Brainstorm questions
- Turn a dashboard alive
- Advanced data science

- Save & download files
- Send an email
- File a ticket

# Data Analysis Prompts Leveraging AI



## Variable Guide: by Data Collection Form

### Enrollment Visit p 1

**Variable Name:** V00COHORT

**Label:** EV:Subcohort assignment (calc)

**SAS Dataset:** Enrollees

**Release Comments:** Notes available: see individual SAS dataset documentation for details

| Category            | SubCategory         |
|---------------------|---------------------|
| Knee pain/OA status | Knee pain/OA status |
| Study eligibility   | Other               |

|                              | Value | N     | % | Cumulative N | Cumulative % |
|------------------------------|-------|-------|---|--------------|--------------|
| 1: Progression               | 1,390 | 28.98 |   | 1,390        | 28.98        |
| 2: Incidence                 | 3,284 | 68.47 |   | 4,674        | 97.46        |
| 3: Non-exposed control group | 122   | 2.54  |   | 4,796        | 100.00       |

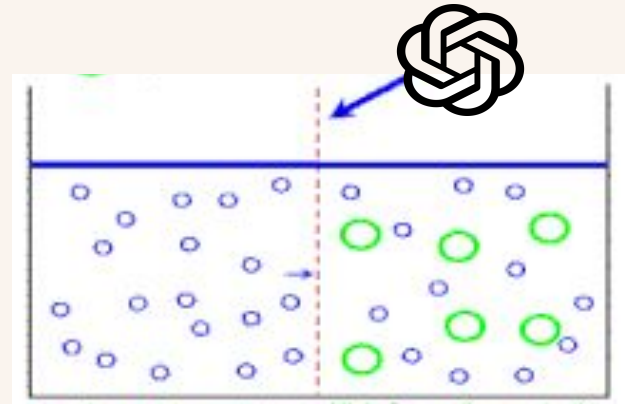
- **Data summarization:** Will perform upon upload.
- **Data prep/cleaning:**
  - a. What are some anomaly checks to run in this dataset?
  - b. Add a column after "V00COHORT" to the enrollees dataset labeled V00COHORT\_Verbatim" and in that column for each subject's ID, enter the text "Progression" if "V00COHORT" is a 1, "Incidence" if "V00COHORT" is a 2, and "Control" if "V00COHORT" is a 3.
- **Data analysis:** Here are two interesting data analytics questions you could ask this dataset:
  - 1. What is the race and ethnicity breakdown for each cohort?
  - 2. How many subjects by id in the "outcomes 99" table had a knee replacement?



# **Future Reactions: Conclusions and Next Steps**

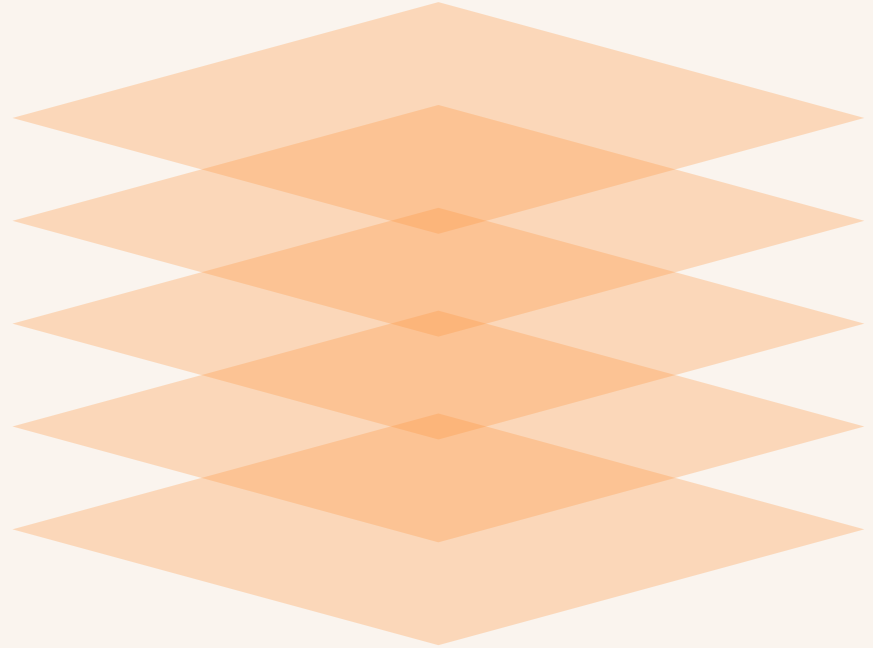
# AI Implementation Tips

- Have a plan for enterprise tool selection, implementation, and training to drive culture and adoption
- Consider data governance, tool limitations, and potential use cases to manage risk
- Helps teams find opportunities for quick wins with lower-risk knowledge and data
- There are many opportunities in clinical operations and data analytics teams where AI can drive efficiency and innovation



# Endless Opportunities for AI in Pharma

- Areas of opportunity:
  - Documentation & TMF
  - Validation & UAT
  - CRF and edit check configuration
  - Data analysis & visualization
  - New vendor capabilities
  - Statistical programming
  - Clinical trial operations
  - and so many more...





# Questions?